# Sampling Distributions for Sample Means – Baseball Salaries

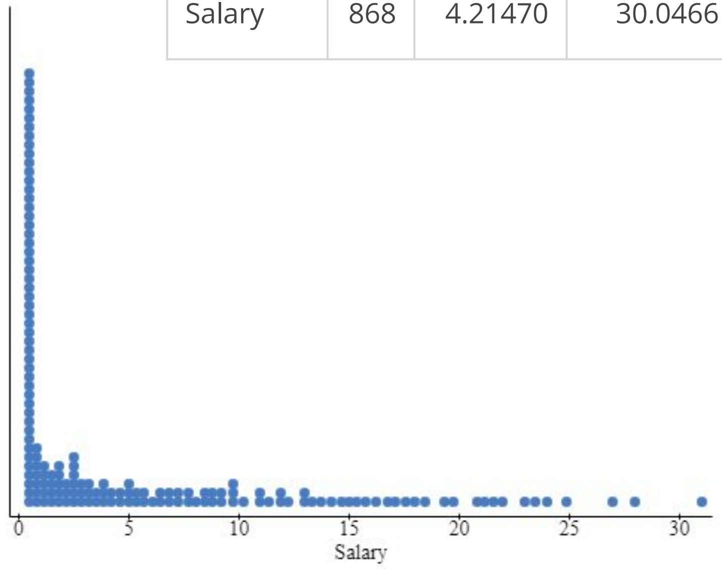
MATH 213

# Group 1

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Salary | 868 | 4.21470 | 30.0466 | 5.48147 | 0.18605 | 1.65 | 30.492 | 0.508 | 31 | 0.519 | 5.7875 |




Write a description of the distribution of baseball players' salaries.

**The distribution of baseball players' salaries is sharply skewed right with the majority of players being between 1-5 million dollars per year. The median player got paid 1.65 million for their salary.**

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.
**4.214 million**

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

This would be a parameter since it is describing the entire population of MLB players

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

The dot being placed represents the average mean of salaries within 30 random samples.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

They are not the same

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 7.288 |
| 2 | 4.436 |
| 3 | 3.884 |
| 4 | 2.622 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

They are not the same and all the samples are not that close to one another. From the samples there is only one of them that is close to the population mean.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot?

**1005**

What does each dot represent?

**An average salary for baseball players**

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**Note / Definition:** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): 4.527

Shape: normal, slightly skewed right

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: 4.525

Shape: normal, slightly skewed right

Standard error: 0.581

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution **decreases**

As the sample size increases, the center of the sampling distribution gets closer and closer to **4.5**

As the sample size increases, the shape of the sampling distribution gets closer and closer to being a normal distribution

# Group 2

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable “salary”, and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
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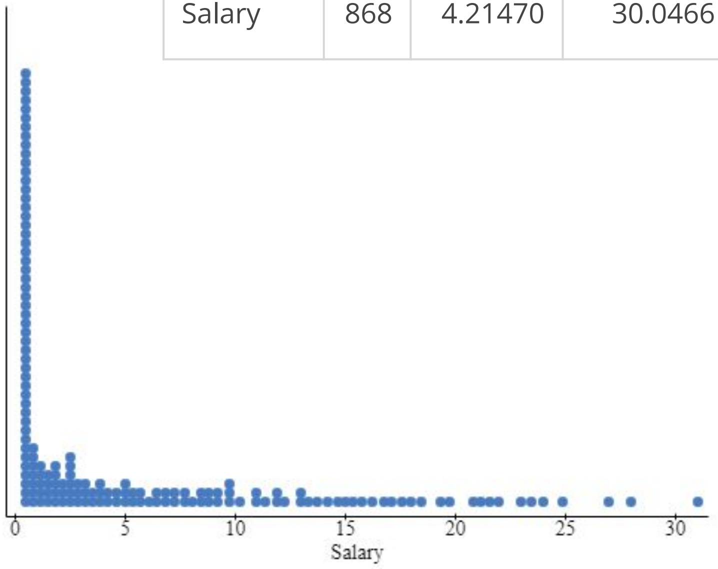
| Salary | 868 | 4.21470 | 30.0466 | 5.48147 | 0.18605 | 1.65 | 30.492 | 0.508 | 31 | 0.519 | 5.7875 |




Write a description of the distribution of baseball players’ salaries.

The plot showing the distribution of baseball players’ salaries has a center of 1.65 million (median), give or take 5.48 million. It is heavily right skewed, which means the data for higher salary is far more spread out while the data for lower salaries are more concentrated. Most of the salaries in the sample are between 0.51 million and 5.79 million.

Even though this is skewed, we’ll focus on mean salary. Write down the mean baseball salary from 2015.

The mean salary is 4.21 million.

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we’ve been talking about a parameter or a statistic?

If the players are considered the population, then the mean salary is a **parameter**.

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

This dot represents the mean salary of a random sample of size 30 from the population.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

No.

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

The overall mean of the sample mean salaries is 4.223, which is pretty close.

| Sample | Sample mean salary |
| --- | --- |
| 1 | 2.019 |
| 2 | 4.577 |
| 3 | 4.595 |
| 4 | 5.7 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

No, the sample means are not all the same. The average sample salary mean is 4.223, with a standard deviation of 1.56, which is pretty close.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? 1004

What does each dot represent? A sample mean from a sample size of 30

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**Note / Definition:** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): 4.536

Shape: The distribution looks normal/bell-shaped

Spread (standard error) : 1.143

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: 4.474

Shape: Looks normal/bell-shaped, centered on a lower salary

Standard error: 0.590

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution decreases

As the sample size increases, the center of the sampling distribution gets closer and closer to the population center

As the sample size increases, the shape of the sampling distribution gets closer and closer to being a normal distribution?

# Group 3

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
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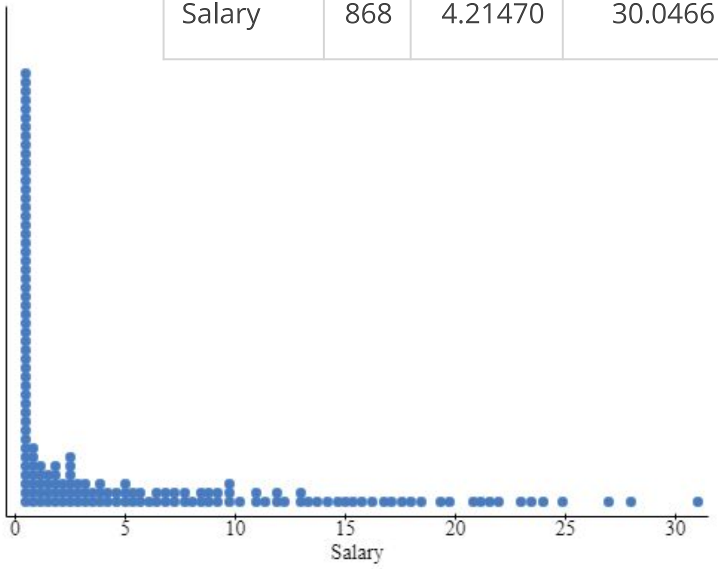


Write a description of the distribution of baseball players' salaries.
The samples are right skewed, with a center around 1.65. Most salaries are at the one million dollar range.

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.
4.1247

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

It is a parameter.

- Go to the StatKey applet at http://www.lock5stat.com/StatKey/index.html
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- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

It represents the mean players yearly salary in the MLB, for a sample size of 30.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 5.744 |
| 2 | 4.49 |
| 3 | 3.419 |
| 4 | 2.591 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

5 - 3.471

6 - 5.206

7 - 4.027

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

They are not the same, they are not super close, the closest being 4.49. But two being very fay away, 2.6 and 5.7.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot?

What does each dot represent?

Each plot is the average salary of a sample of 30 MLB players.

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**<u>Note / Definition:</u>** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): The center of the data is 4.258.

Shape: The shape is relatively bell shaped, with an even distribution on each side of the mean.

Spread (standard error) : The spread of the data ranges with an standard deviation of 5.481

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: The center of the data is 4.208.

Shape: Uniform distribution

Standard error: The standard deviation of 5.401

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution remains equal.

As the sample size increases, the center of the sampling distribution gets closer and closer to

As the sample size increases, the shape of the sampling distribution gets closer and closer to being a bell-curve.

# Group 4

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
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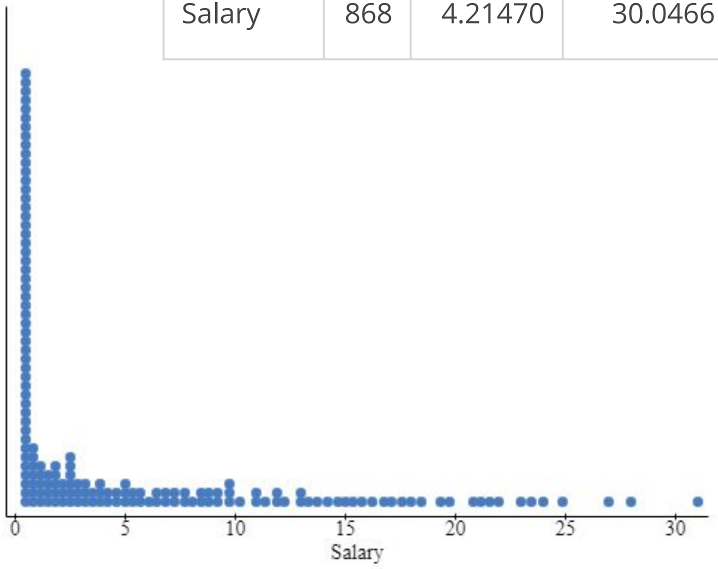


Write a description of the distribution of baseball players' salaries.

The dotplot of the salaries for the 868 plays in the MLB is right skewed, meaning fewer players had salaries in the higher millions and the majority of players had salaries in the lower millions.

Even though this is skewed, we’ll focus on mean salary. Write down the mean baseball salary from 2015.

4.21

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we’ve been talking about a parameter or a statistic?

The mean salary is a parameter because it is the mean of the entire population.

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
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**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

The average salary for a sample of 30 baseball players.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

No

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 4.602 |
| 2 | 4.773 |
| 3 | 4.183 |
| 4 | 4.871 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

They are not all the same, but they are close to one another; all four samples are within about .7 of one another. All four samples are within about .6 of the population mean salary.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? 1009

What does each dot represent? Average salary of a sample of 30 baseball players

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

<u>**Note / Definition:**</u> The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): The typical sample of 30 MBL players have an average salary of 4.311 million

Shape: Normal

Spread (standard error) : 0.982

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: The typical sample of 100 MBL players have an average salary of 4.198 million

Shape: Normal

Standard error: 0.504

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution decreases.

As the sample size increases, the center of the sampling distribution gets closer and closer to the population parameter

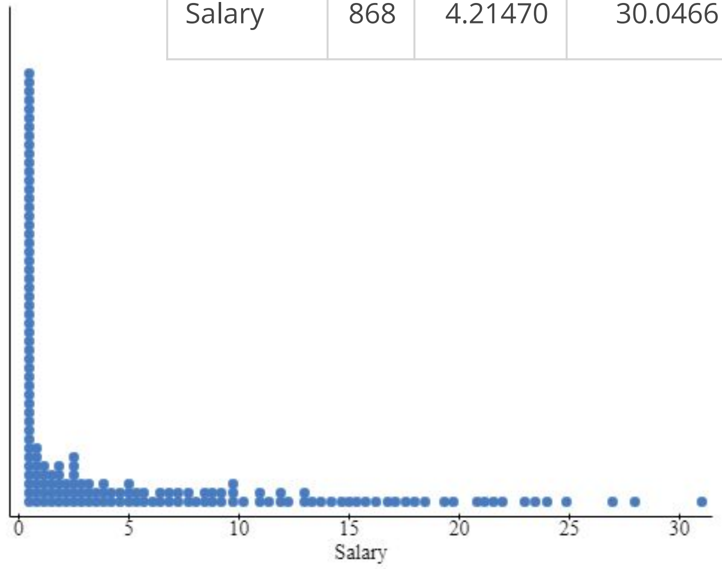
As the sample size increases, the shape of the sampling distribution gets closer and closer to being a bell curve

# Group 5

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Salary | 868 | 4.21470 | 30.0466 | 5.48147 | 0.18605 | 1.65 | 30.492 | 0.508 | 31 | 0.519 | 5.7875 |




Write a description of the distribution of baseball players' salaries.

The dotplot of salaries for 868 active MLB players is right skewed, meaning few players earn more than 1.65 million dollars for the 2015 season, with most players earning between 0.519 and 5.7875 million dollars.

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.
4.21470

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

Parameter, since the mean is representative of the average of all active MLB players in the 2015 season.

- Go to the StatKey applet at http://www.lock5stat.com/StatKey/index.html
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

The dot represents the average salary for the sample of 30 MLB players.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

no

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 5.551 |
| 2 | 4.834 |
| 3 | 4.794 |
| 4 | 4.653 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

Mean: 4.21470

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

The values are relatively close to one another, and they are within a million dollar range of the mean.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? **1000**

What does each dot represent? **The mean salary for the 30 players in our samples**

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread

<u>**Note / Definition:**</u> The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): 4.222

Shape: mostly normal, slight right skew

Spread (standard error): 0.971

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: 4.223

Shape: normal

Standard error: 0.512

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution decreases

As the sample size increases, the center of the sampling distribution gets closer and closer to 4.21470, the mean

As the sample size increases, the shape of the sampling distribution gets closer and closer to being a normal distribution.

# Group 6

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable “salary”, and a table of statistics for that variable.

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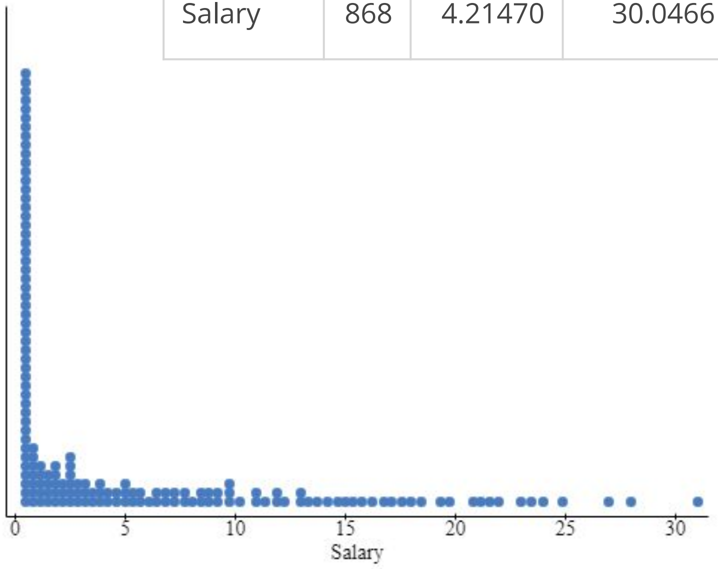


Write a description of the distribution of baseball players’ salaries.

The players have a very wide range of salary with a median salary of $1.65M. Half of the players have a salary between $519 thousand and $5.79M while the highest paid player is earning a salary of $31M and the lowest paid is earning $508 thousand, just a small bit outside the median salary range.

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.

4.21470

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

The mean salary is a parameter of the 868 members of the MLB population.

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
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**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

The dot is representative of the mean salary of the 30 salary sample.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

No.

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about? **It is not the same and but they are in the rough ballpark.**

| Sample | Sample mean salary |
| --- | --- |
| 1 | 3.49 |
| 2 | 4.126 |
| 3 | 7.654 |
| 4 | 4.771 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

They are not the same but they are, for the most part, still close to each other centering around the 4.215 mean of the population.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? **1011 dots**
What does each dot represent? **The mean salary of 30 simulated samples.**
This dotplot is called the ***sampling distribution for sample means.***
Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

<u>**Note / Definition:**</u> The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): 4.254

Shape: Shape appears to have a small right skew with the highest value being 7.932

Spread (standard error) : 0.960

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: 4.160

Shape:  The shape is very flat with a clear right skew.

Standard error: 0.925

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution **stabilizes around a given number of roughly .98.**

As the sample size increases, the center of the sampling distribution gets closer and closer to **4.215**

As the sample size increases, the shape of the sampling distribution gets closer and closer to being a **normal distribution curve.**

# Group 7

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

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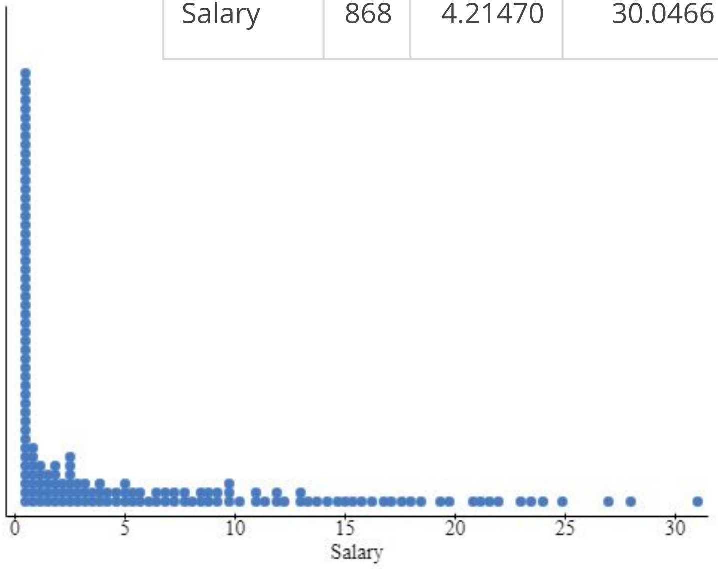



Write a description of the distribution of baseball players' salaries.

The graph shows a description of the salaries of 868 baseball players in 2015. The graph is skewed right, with the average salary being 4.2 million. 50% of players salary is between 0.5 million - 5.8 million. The range of salaries ranges from 0.5 million - 31 million.

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.
Mean salary (millions of dollars): 4.21470

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

stastic

- Go to the StatKey applet at http://www.lock5stat.com/StatKey/index.html
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

It represents the average salary of a sample of 30 baseball players

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

No

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 4.845 |
| 2 | 3.961 |
| 3 | 1.782 |
| 4 | 0.787 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

The data points are relatively similar given the spread of the data. Considering the data has a spread of 30 million, the samples are close to the mean.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? 1000

What does each dot represent? Mean salary of a sample of 30 baseball players.

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**<u>Note / Definition:</u>** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): 4.223

Shape: Normal

Spread (standard error) : 1.012

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: 4.167

Shape: flat

Standard error: 5.481

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution decreases

As the sample size increases, the center of the sampling distribution gets closer and closer to the mean

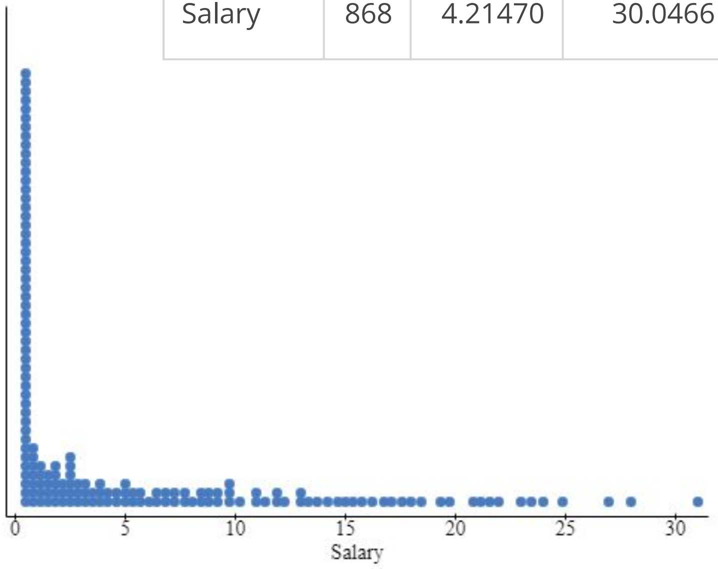
As the sample size increases, the shape of the sampling distribution gets closer and closer to being a normal bell shape

# Group 8

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Salary | 868 | 4.21470 | 30.0466 | 5.48147 | 0.18605 | 1.65 | 30.492 | 0.508 | 31 | 0.519 | 5.7875 |




Write a description of the distribution of baseball players' salaries.
**A majority of baseball players in 2015 had a salary of at least $1 million. The salaries aren't that varied between the players, but there is certainly a wide range of salaries that are being paid. The distribution of salaries itself is extremely skewed to the right, with most of the salaries resting at around $1 million.**

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.
**~$4.21 million**

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

**The mean salary in this example is a statistic, this is because the mean is data taken from the entire population of the mlb, as opposed to a smaller sample.**

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

**This dot represents the mean of 50 randomly selected samples from the plot.**

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

**No the sample mean salaries are different almost every time**

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 4.393 |
| 2 | 6.359 |
| 3 | 7.221 |
| 4 | 3.158 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

**They are not all the same and they are not relatively close to each other. One of the samples is close to the mean salary however the rest aren’t very close to the mean.**

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? **1000**

What does each dot represent? **The mean of 50 random data points from the graph**

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**Note / Definition:** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): **4.246**

Shape: **bell with no skew**

Spread (standard error) : **0.961**

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: **4.208**

Shape: **bell with no skew**

Standard error: **0.522**

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution **decreases**

As the sample size increases, the center of the sampling distribution gets closer and closer to **the mean of the data using the whole population**

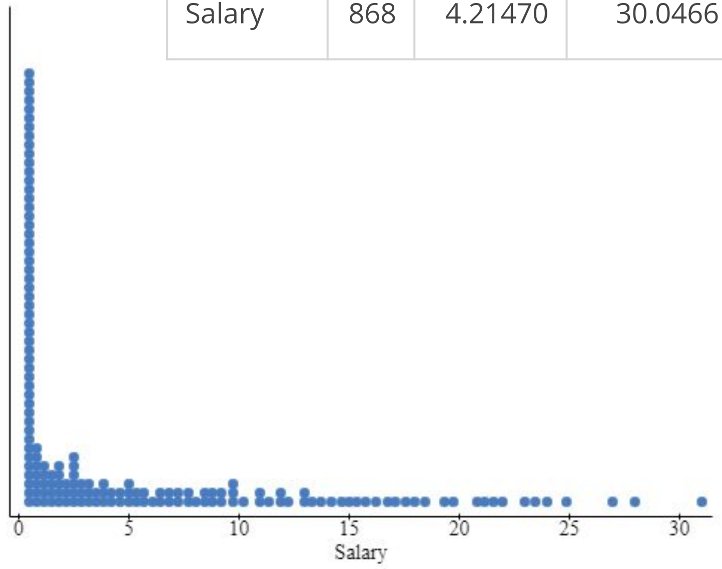
As the sample size increases, the shape of the sampling distribution gets closer and closer to being a **perfect bell**

# Group 9

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Salary | 868 | 4.21470 | 30.0466 | 5.48147 | 0.18605 | 1.65 | 30.492 | 0.508 | 31 | 0.519 | 5.7875 |




Write a description of the distribution of baseball players' salaries.
This dotplot represents the distribution of salaries between players in major league baseball teams in 2015. The graph is a heavily right skewed shape, with the middle point of the data being 1.65 million, with most of the points landing between .519 mill and 5.7875 mill.

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

4.2

It's a parameter, because this suggests that our sample is in fact the population, so the statistic would then be a parameter.

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

The average mean of salaries out of random samples of 30 from the baseball players.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

no

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about? No, and the closest one is about .5 or more apart.

| Sample | Sample mean salary |
|---|---|
| 1 | 2.642 |
| 2 | 3.067 |
| 3 | 3.727 |
| 4 | 4.917 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary?

They are not all the same, they’re at most about 1-.5 apart from each other. And the one that is closest to the population mean is .5 away from it.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? 1005

What does each dot represent? The average means of random samples of 30

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**Note / Definition:** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): 4.167

Shape: slightly right skewed

Spread (standard error) : 1.009

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: 4.205

Shape: normal

Standard error: .516

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution decreases

As the sample size increases, the center of the sampling distribution gets closer and closer to the population mean

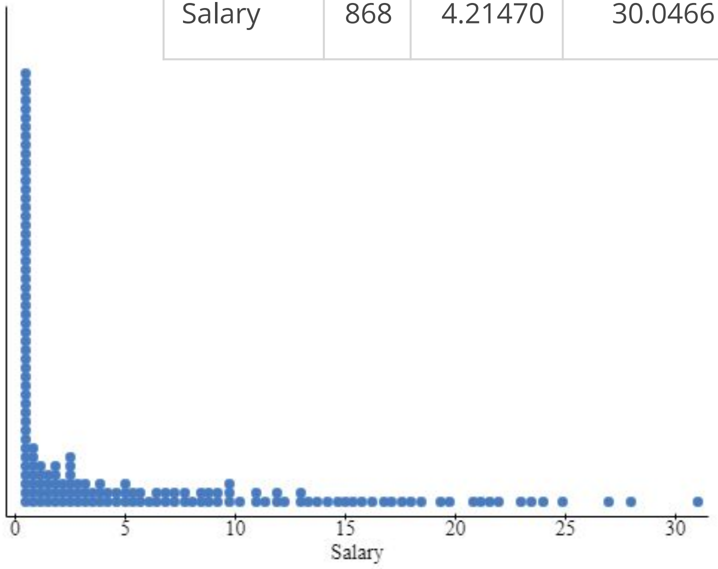
As the sample size increases, the shape of the sampling distribution gets closer and closer to being a normal bell curve

# Group 10

The dataset BaseballSalaries2015 shows the salary (in millions of dollars) for all 868 players who were on the active roster for Major League Baseball teams at the start of the 2015 season. The file also shows the team and primary position for each player. Here is a link to the codebook for the dataset:

Below, see a graph of the variable "salary", and a table of statistics for that variable.

| Column | n | Mean | Variance | Std. dev. | Std. err. | Median | Range | Min | Max | Q1 | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Salary | 868 | 4.21470 | 30.0466 | 5.48147 | 0.18605 | 1.65 | 30.492 | 0.508 | 31 | 0.519 | 5.7875 |




Write a description of the distribution of baseball players' salaries.

This graph shows the salary of all the active players in the MLB from the 2015 season. The graph seems to be right skewed, with most players salary being at 1.6. The deviation of how much a player gets paid is 0.186.

Even though this is skewed, we'll focus on mean salary. Write down the mean baseball salary from 2015.
4.214

For the purpose of this activity, we will consider the players in the 2015 dataset to be **the population of all MLB players at that time**. With this in mind, is the mean salary we've been talking about a parameter or a statistic?

I would say that the mean in this graph is a parameter. This is because the mean is using all the MLB players in the league which is a population which for a whole population this is the average of what a player is making in the league.

- Go to the StatKey applet at [http://www.lock5stat.com/StatKey/index.html](http://www.lock5stat.com/StatKey/index.html)
- Click “Mean” under “Sampling Distributions.”
- In the upper left corner, choose the “Baseball Players–2e” data set.

Notice once the data has populated that there is a Data Plot in the top right of the population of all salaries, along with the associated descriptive statistics. This information should match what we saw This information should match what we saw in the graph and table on the first slide. Check this to make sure.

**Here, we will imagine for a while that we don’t know the population mean salary and pretend that we are using sample means to predict the population mean.**

We will tell StatKey to select a random sample of 30 salaries from the population of all salaries. To do this, make sure that 30 is entered in the blank for “Choose samples of size ___”. Then click Generate One Sample.

When you click Generate One Sample, you will notice that StatKey places a dot on the dotplot. What does this dot represent? (Be specific.)

This dot represents a random sample of 30 salaries from the population that a player has in the MLB and it was randomly selected from the 30 salaries that from our graph.

Click “Generate One Sample” a few more times.

Are the sample mean salaries the same as each other every time?

No they are not, the salaries are spread out and not very concentrated.

Are the sample mean salaries the same as the population mean of 4.215 million dollars? How close are they, about?

| Sample | Sample mean salary |
|---|---|
| 1 | 4.472 |
| 2 | 4.102 |
| 3 | 3.392 |
| 4 | 5.634 |

7. Click Generate One Sample three more times: have StatKey select three more samples. Record each new sample mean in the table above for a total of four sample means.

8. Consider the sample means for the four samples. Are they all the same? Are they close to one another? How close are they to the population mean salary? We have two points that are close to the mean salary which is 4.125 million. But we do have some outliers which are the 3.392 and the 5.634.

Next, click Generate 1000 Samples.

How many dots are now on your dotplot? 1010

What does each dot represent? Each dot represenets a mean of a random 30 salaries selected from our population of salaries

This dotplot is called the ***sampling distribution for sample means.***

Stare at the sampling distribution and jot some notes relating to its center, shape, and spread.

**Note / Definition:** The standard deviation of a sampling distribution is given a special name – it is called the **standard error**. Remember, it tells us, approximately how far, on average, a data point is from the mean.

Center (mean): The average of the salaries is 4.159 which is the population proportion

Shape: The graph seems to be bell shaped, there is not any significant skew

Click Reset Plot.

Change the sample size to 100.

Have StatKey generate a new sampling distribution by Generate 1000 Samples.

What is the

Center: The average of salaries for the sample size of 100 is 4.208.

Shape: The shape is also bell shaped without any skew

Standard error: The standard error for the sample size 100 is 0.531 meaning on average a dot that is that much from the mean.

Fill in the blanks with your observations from the previous slide.

As the sample size increases, the standard error of the sampling distribution decreases

As the sample size increases, the center of the sampling distribution gets closer and closer to 4.125

As the sample size increases, the shape of the sampling distribution gets closer and closer to being a bell shaped graph